

## **Chapter 1: Foundations of Machine Learning**

Machine learning is a subfield of artificial intelligence that enables systems to learn patterns from data without being explicitly programmed. The fundamental premise is that algorithms can improve their performance through experience.

## Chapter 1 (continued)

Supervised learning requires labeled training data, while unsupervised learning discovers hidden structures. Reinforcement learning agents optimize long-term rewards through trial and error interactions with their environment.

## Chapter 3: Distributed Training

As neural networks grew larger, researchers developed distributed training techniques to parallelize computation across multiple GPUs. This breakthrough enabled training of models with billions of parameters on commodity hardware clusters.

## Chapter 3 (continued)

The transformer architecture revolutionized natural language processing by introducing self-attention mechanisms. This innovation allowed models to capture long-range dependencies without recurrent connections.